

hw4-proofs-sets-recursion: Sample Solutions

CSE20W26

In this assignment, you will analyze statements and determine if they are true or false using valid proof strategies. You will also determine if candidate arguments are valid. You will work with recursively defined sets and functions and prove properties about them, practicing induction and other proof strategies.

Relevant class material: Weeks 5, 6, and 7.

You will submit this assignment via Gradescope (<https://www.gradescope.com>) in the assignment called “hw4-proofs-sets-recursion”.

In your proofs and disproofs of statements below, justify each step by reference to a component of the following proof strategies we have discussed so far, and/or to relevant definitions and calculations.

Assigned questions

1. Number properties. Consider the predicate $F((a, b)) = “a \text{ is a factor of } b”$ over the domain $\mathbb{Z}^{\neq 0} \times \mathbb{Z}$. Consider the following quantified statements

- | | |
|--|---|
| (i) $\forall x \in \mathbb{Z} (F((1, x)))$ | (v) $\forall x \in \mathbb{Z}^{\neq 0} \exists y \in \mathbb{Z} (F((x, y)))$ |
| (ii) $\forall x \in \mathbb{Z}^{\neq 0} (F((x, 1)))$ | (vi) $\exists x \in \mathbb{Z}^{\neq 0} \forall y \in \mathbb{Z} (F((x, y)))$ |
| (iii) $\exists x \in \mathbb{Z} (F((1, x)))$ | (vii) $\forall y \in \mathbb{Z} \exists x \in \mathbb{Z}^{\neq 0} (F((x, y)))$ |
| (iv) $\exists x \in \mathbb{Z}^{\neq 0} (F((x, 1)))$ | (viii) $\exists y \in \mathbb{Z} \forall x \in \mathbb{Z}^{\neq 0} (F((x, y)))$ |

- (a) Which of the statements (i) - (viii) is being **proved** by the following proof:

By universal generalization, **choose** e to be an **arbitrary** integer. We need to show that $F((1, e))$. By definition of the predicate F , we can rewrite this goal as $\exists c \in \mathbb{Z} (e = c \cdot 1)$. We pick the **witness** $c = e$, which is an integer and therefore in the domain. Calculating, $c \cdot 1 = e \cdot 1 = e$, as required. Since the predicate $F(1, e)$ evaluates to true for the arbitrary integer e , the claim has been proved. $\square$

Hint: it may be useful to identify the key words in the proof that indicate proof strategies.

Solution: From the keywords “By universal generalization”, “arbitrary e ,” we know that the proof is trying to prove a universal quantification by universal generalization, so the statement should start with $\forall e$.

From the keyword “integer,” we know the domain is the set of integers $\mathbb{Z}$.

From the next sentence “We need to show that $F((1, e))$,” we know that the predicate is $F((1, e))$. The rest are just evaluating $F((1, e))$.

When combined together, the statement is $\forall e \in \mathbb{Z} F((1, e))$, which is equivalent to (i).

- (b) Which of the statements (i) - (viii) is being **disproved** by the following proof:

To disprove the statement, we need to find a counterexample. We choose 2, a nonzero integer so in the domain. We need to show that $\neg F((2, 1))$. By definition of the predicate F , we can rewrite this goal as $1 \bmod 2 \neq 0$. By definition of integer division, since $1 = 0 \cdot 2 + 1$ (and $0 \leq 1 < 2$), $1 \bmod 2 = 1$ which is nonzero so the counterexample works to disprove the original statement. $\square$

Hint: it may be useful to identify the key words in the proof that indicate proof strategies.

Solution: From the keywords “find a counterexample.” we infer that the proof is trying to disprove a universal quantification by counterexample, so the statement should start with $\forall e$. The counterexample e the proof picked is $e = 2$.

From the keyword “nonzero integer so in the domain,” we infer the domain is the set of nonzero integers $\mathbb{Z}^{\neq 0}$.

From the next sentence “We need to show that $\neg F((2, 1))$,” we infer that the predicate is $F((e, 1))$ (as $e = 2$). The rest are just evaluating $F((2, 1))$.

When combined together, the statement is $\forall e \in \mathbb{Z}^{\neq 0} F((e, 1))$, which is equivalent to (ii).

- (c) Translate the statement to English, state whether is it true or false, and then justify your answer (by proving the statement or its negation). (*Graded for correctness of translation and evaluation of statement (is it true or false?) and fair effort completeness of the proof*)

$$\exists x \in \mathbb{Z}^{\neq 0} \exists y \in \mathbb{Z}^{\neq 0} (\neg(x = y) \wedge F((x, y)) \wedge F((y, x)))$$

Solution: The English statement is “There exists a nonzero integer x and a nonzero integer y such that $x \neq y$, x is a factor of y , and y is a factor of x ”. This statement is true. Consider the witness $x = 2, y = -2, 2 \neq -2$. Since $2 = (-2) \cdot (-1)$ and $-2 = 2 \cdot (-1)$, $(\neg(x = y) \wedge F((x, y)) \wedge F((y, x)))$ is true.

- (d) Translate the statement to English, state whether is it true or false, and then justify your answer (by proving the statement or its negation). (*Graded for correctness of translation and evaluation of statement (is it true or false?) and fair effort completeness of the proof*)

$$\forall x \in \mathbb{Z}^{\neq 0} \forall y \in \mathbb{Z}^{\neq 0} (F((x, y)) \rightarrow \neg F((y, x)))$$

Solution: The English statement is “For all nonzero integer x and for all nonzero integer y , if x is a factor of y , then y is not a factor of x ”. This statement is false. Consider the same $x = 2, y = -2$ from (c) as a counterexample. Since $2 = (-2) \cdot (-1)$ and $-2 = 2 \cdot (-1)$, $F((x, y))$ is true and $\neg F((y, x))$ is false. Hence, $(F((x, y)) \rightarrow \neg F((y, x)))$ is false.

- (e) Translate the statement to English, state whether is it true or false, and then justify your answer (by proving the statement or its negation). (*Graded for correctness of translation and evaluation of statement (is it true or false?) and fair effort completeness of the proof*)

$$\exists x \in \mathbb{Z}^{\neq 0} \exists y \in \mathbb{Z} (F((x, y)) \wedge F((x + 1, y)) \wedge F((x + 2, y)))$$

Solution: The English statement is “There exists a nonzero integer x and an integer y such that $x, x + 1$, and $x + 2$ are all factors of y ”. This statement is true. Consider the witness $x = 1, y = 6$. 1, 2, and 3 are all factors of 6, since $6 = 1 \cdot 6 = 2 \cdot 3 = 3 \cdot 2$ therefore $F((x, y)) \wedge F((x + 1, y)) \wedge F((x + 2, y))$ is true.

- (f) Translate the statement to English, state whether is it true or false, and then justify your answer (by proving the statement or its negation). (*Graded for correctness of translation and evaluation of statement (is it true or false?) and fair effort completeness of the proof*)

$$\forall x \in \mathbb{Z}^{\neq 0} (F((x, x^2)) \wedge F((x, x^3)))$$

Solution: The English statement is “each nonzero integers are a factor of its square and its cube”. This statement is true. By universal generalization, let x be a nonzero integer. $F((x, x^2))$ is true because $x \cdot x = x^2$ so the integer x witnesses that x is a factor of x^2 . $F((x, x^3))$ is true because $x \cdot x^2 = x^3$ so the integer x^2 witnesses that x is a factor of x^3 . Since $F((x, x^2)) \wedge F((x, x^3))$ is true for all x , the claim has been proven.

2. Structural induction. Recall that we define the set of bases as $B = \{\mathbf{A}, \mathbf{C}, \mathbf{U}, \mathbf{G}\}$. The set of RNA strands S is defined (recursively) by:

$$\begin{array}{ll} \text{Basis Step:} & \mathbf{A} \in S, \mathbf{C} \in S, \mathbf{U} \in S, \mathbf{G} \in S \\ \text{Recursive Step:} & \text{If } s \in S \text{ and } b \in B, \text{ then } sb \in S \end{array}$$

where sb is string concatenation. The function $rnalen$ that computes the length of RNA strands in S is defined recursively by $rnalen : S \rightarrow \mathbb{Z}^+$

Basis step: If $b \in B$ then $rnalen(b) = 1$

Recursive step: If $s \in S$ and $b \in B$, then $rnalen(sb) = 1 + rnalen(s)$

The function *basecount* that computes the number of a given base b appearing in a RNA strand s is defined recursively:

$$\begin{array}{lll} & \text{basecount} : S \times B & \rightarrow \mathbb{N} \\ \text{Basis Step:} & \text{If } b_1 \in B, b_2 \in B & \text{basecount}((b_1, b_2)) = \begin{cases} 1 & \text{when } b_1 = b_2 \\ 0 & \text{when } b_1 \neq b_2 \end{cases} \\ \text{Recursive Step:} & \text{If } s \in S, b_1 \in B, b_2 \in B & \text{basecount}((sb_1, b_2)) = \begin{cases} 1 + \text{basecount}((s, b_2)) & \text{when } b_1 = b_2 \\ \text{basecount}((s, b_2)) & \text{when } b_1 \neq b_2 \end{cases} \end{array}$$

- (a) Translate the statement to English, state whether is it true or false, and then justify your answer (by proving the statement or its negation). (*Graded for correctness of translation and evaluation of statement (is it true or false?) and fair effort completeness of the proof*)

$$\forall b \in B \exists s \in S (\text{rinalen}(s) = \text{basecount}((s, b)))$$

Solution: The English statement is “For all base b , there exists a RNA strand s such that the length of s equals the number of b appearing in s ”. This statement is true. By universal generalization, let b be an arbitrary base. Consider $s = b$ be a witness, which is in the domain as $b \in B$ and the basis step in the definition of S ensures $b \in S$. By applying the basis step of the two functions *basecount* and *rinalen*, we have $\text{rinalen}(s) = \text{basecount}((s, b)) = 1$.

- (b) Translate the statement to English, state whether is it true or false, and then justify your answer (by proving the statement or its negation). (*Graded for correctness of translation and evaluation of statement (is it true or false?) and fair effort completeness of the proof*)

$$\forall s \in S (\text{rinalen}(s) > \text{basecount}((s, A)))$$

Solution: The English statement is “For all RNA strand s , the length of s is greater than the number of A appearing in s ”. This statement is false. Consider a counterexample $s = A$, which is in the domain by the basis step in the definition of S . By applying the basis step of the two functions *basecount* and *rinalen*, we have $\text{rinalen}(s) = \text{basecount}((s, A)) = 1$, which implies $(\text{rinalen}(s) > \text{basecount}((s, A)))$ is false.

- (c) Translate the statement to English, state whether is it true or false, and then justify your answer (by proving the statement or its negation). (*Graded for correctness of translation and evaluation of statement (is it true or false?) and fair effort completeness of the proof*)

$$\forall s \in S \forall b \in B (\text{basecount}((s, b)) > 0)$$

Solution: The English statement is “For all RNA strand s and for all base b , the number of b appearing in s is greater than 0”. This statement is false. Consider a counterexample $s = A$ and $b = C$, which are both in the domain by the basis step in the definition of S and the set roster of B . By applying the basis step of function *basecount*, we have $\text{basecount}((s, b)) = \text{basecount}((A, C)) = 0$, which implies $(\text{basecount}((s, b)) > 0)$ is false.

3. Games and induction. The game of Nim-Var is a two-player game (which is a variant of Nim). At the start of the game, there are two piles, each containing n jelly beans (n is a positive integer). On a player's turn, that player picks one of the two piles and does **one** of the following: either

- removes some positive number of jelly beans from that pile, **or**
- moves some positive number of jelly beans (that is less than the current total number of jelly beans in the pile) from that pile to the other. *Note: if there is only one jelly bean left in a pile, the player cannot move this jelly bean to the other pile.*

The player to take the last jelly bean wins. Use strong induction to prove that the second player always has a winning strategy in Nim-Move. A complete and correct solution will first identify what the strategy is, and then prove that following this strategy will lead the second player to win the game (no matter what the first player chooses to do at each turn).

Sample Solution:

Strategy for player 2: If player 1 chooses to remove k jelly beans from a pile, then we also remove k jelly beans from the opposite pile. If player 1 chooses to move jelly beans (e.g. if they move k jelly beans from pile A to pile B), then we remove $2k$ jelly beans from pile B.

Proof by strong induction: WTS: for all positive integers n , Player 2's strategy guarantees a win in the game that starts with n jellybeans in each pile.

Basis step: WTS Player 2's strategy guarantees a win when each pile starts with 1 jellybean. In this case, Player 1 has to remove the jellybean from one of the piles (because they can't take from both piles at once and they can't move jelly beans). Following the strategy, Player 2 removes the jellybean from the other pile, and wins because this is the last jellybean.

Recursive step: Let n be a positive integer. As the strong induction hypothesis, assume that Player 2's strategy guarantees a win in the games where there are $1, 2, \dots, n$ many jellybeans in each pile at the start of the game.

WTS that Player 2's strategy guarantees a win in the game where there are $n + 1$ jellybeans in each pile at the start of the game.

In this game, the player 1's action can be categorized into these 3 cases: [moves k jelly beans from a pile to the other where $1 \leq k \leq n$; removes all $n + 1$ jelly beans from a pile; removes c jelly beans from a pile where $1 \leq c \leq n$].

Case 1: If player 1 moves k jelly beans from pile A to the other pile B where $1 \leq k \leq n$, then at this time, pile A has $n + 1 - k$ jelly beans and pile B has $n + 1 + k$ jelly beans. Based on our strategy, we will remove $2k$ jelly beans from pile B, which leaves it with $n + 1 + k - 2k = n + 1 - k$ jelly beans. Since $1 \leq k \leq n$, this number $(n + 1) - k$ is also between 1 and n . Thus, at this stage of the game, the game appears identical to a new game where the two piles have an equal number of jellybeans between 1 and n . Thus, the strong induction hypothesis applies, and Player 2's strategy guarantees they win.

Case 2: If player 1 removes a whole pile of $n + 1$ jelly beans, then in their first move, Player 2 wins when following their strategy of removing the same number of jelly beans from the other pile because they remove all of the second pile, which includes the last jelly bean.

Case 3: If player 1 removes c jelly beans from a pile where $1 \leq c \leq n$, then after Player 2's first move following the strategy, the two piles have an equal number of $(n + 1) - c$ jelly beans. Since $1 \leq c \leq n$, this number $(n + 1) - c$ is also between 1 and n . Thus, at this stage of the game, the game appears identical to a new game where the two piles have an equal number of jellybeans between 1 and n . Thus, the strong induction hypothesis applies, and Player 2's strategy guarantees they win.

4. Linked Lists. Recall the recursive definition of the set of linked lists of natural numbers (from class)

Basis Step: $[] \in L$

Recursive Step: If $l \in L$ and $n \in \mathbb{N}$, then $(n, l) \in L$

and the definitions of the function which gives the length of a linked list of natural numbers $length : L \rightarrow \mathbb{N}$

Basis Step: $length([]) = 0$

Recursive Step: If $l \in L$ and $n \in \mathbb{N}$, then $length((n, l)) = 1 + length(l)$

the function $append : L \times \mathbb{N} \rightarrow L$ that adds an element at the end of a linked list

Basis Step: If $m \in \mathbb{N}$ then $append([], m) = (m, [])$

Recursive Step: If $l \in L$ and $n \in \mathbb{N}$ and $m \in \mathbb{N}$, then $append((n, l), m) = (n, append(l, m))$

and the function $prepend : L \times \mathbb{N} \rightarrow L$ that adds an element at the front of a linked list

$$prepend(l, n) = (n, l)$$

Sample response that can be used as reference for the detail expected in your answer: To evaluate the result of $length((4, (2, (7, []))))$ we calculate:

$$length((4, (2, (7, [])))) = 1 + length((2, (7, [])))$$

using recursive step of definition of $length$, with $n = 4, l = (2, (7, []))$

$$= 1 + 1 + length((7, []))$$

using recursive step of definition of $length$, with $n = 2, l = (7, [])$

$$= 1 + 1 + 1 + length([])$$

using recursive step of definition of $length$, with $n = 7, l = []$

$$= 1 + 1 + 1 + 0$$

using basis step of definition of $length$, since input to $length$ is $[]$

In this question, we'll consider the combination of these functions with a new function, one that removes the element at the front of the list (if there is any). We define $remove : L \rightarrow L$ by

Basis Step: $remove([]) = []$

Recursive Step: If $l \in L$ and $n \in \mathbb{N}$, then $remove((n, l)) = l$

- (a) What is the result of $remove(append(prepend(((10, []) , 5)), 20))$? For full credit, include all intermediate steps with brief justifications for each.

Solution: We first break down the nested functions into steps:

- i. $a = (10, [])$
- ii. $b = prepend((a, 5))$
- iii. $c = append((b, 20))$
- iv. $d = remove(c)$

Step 1: $b = prepend((a, 5)) = prepend(((10, []), 5)) = (5, (10, []))$

Step 2:

$$\begin{aligned} c = append((b, 20)) &= append(((5, (10, [])), 20)) \\ &= (5, append(((10, []), 20))) \end{aligned}$$

using recursive step of definition of $append$, with $n = 5, l = (10, []), m = 20$

$$= (5, (10, append(([], 20))))$$

using recursive step of definition of $append$, with $n = 10, l = [], m = 20$

$$= (5, (10, (20, [])))$$

using basis step of definition of $append$, since input m to $append$ is 20

Step 3:

$$\begin{aligned} d = remove(c) &= remove((5, (10, (20, [])))) \\ &= (10, (20, [])) \end{aligned}$$

using recursive step of definition of $remove$, with $l = (10, (20, [])), n = 5$

- (b) Prove the statement

$$\forall l \in L (remove(prepend((l, 0))) = l)$$

Sample Solution:

Proof by universal generalization: Let l be an arbitrary linked list $\in L$. By the definition of $prepend$, $prepend((l, 0)) = (0, l)$. Since $(0, l) \neq []$, $remove((0, l))$ will fall into the recursive step and will result in l using recursive step of definition of $remove$, with $l = l, n = 0$. Now we have shown that for any arbitrary $l \in L$, $remove(prepend((l, 0))) = l$. Hence, $\forall l \in L (remove(prepend((l, 0))) = l)$.

(c) Disprove the statement

$$\forall l \in L (\text{remove}(\text{append}(l, 0)) = l)$$

Sample Solution:

Consider the counterexample $l = (1, [])$, which is in L by applying the recursive step in the definition of L once. Now we evaluate $\text{append}(l, 0) = \text{append}((1, []), 0) = (1, \text{append}([], 0)) = (1, (0, []))$ by applying the recursive step once (with $n = 1, l = [], m = 0$) and the basis step once in the definition of function append . Now we evaluate $\text{remove}(\text{append}(l, 0)) = \text{remove}((1, (0, []))) = (0, [])$ by applying the recursive step once (with $l = (0, []), n = 1$) in the definition of function remove . Since $\text{remove}(\text{append}(l, 0)) = (0, []) \neq (1, []) = l$, the universal statement is false.

5. Primes, divisors, and proof strategies. For each statement below, identify the **main logical structure or connective** of the statement, list the proof strategies that could be used to prove and to disprove a statement with that structure, then identify whether the statement is true or false and justify with a proof of the statement or its negation. (*Graded for correctness of identification of logical structure and proof strategies and evaluation of statement (is it true or false?) and fair effort completeness of the proof*)

Sample response that can be used as reference for the detail expected in your answer:

Consider the statement: There is a greatest negative integer.

The main logical structure for this statement is that it is an **existential** statement, as we can see by translating it to symbols:

$$\exists g \in \mathbb{Z}^- \forall x \in \mathbb{Z}^- (g \geq x)$$

To prove an existential statement, the main proof strategy we could use is to find a witness. Proof by cases and proof by contradiction could also be used, because both can be used to prove any statement (no matter its logical structure).

To disprove an existential statement, we would need to prove its negation, which (using DeMorgan's Laws) can be written as a universal statement. Therefore, to disprove this statement the strategies we could use are universal generalization or structural induction (because $\mathbb{Z}^-$ is a recursively defined set) or proof by cases or proof by contradiction. Notice that proof by exhaustion is not possible because the domain is not finite.

The statement is true, as we can see from the witness $g = -1$, since it is in the domain $\mathbb{Z}^-$ and when we evaluate

$$\forall x \in \mathbb{Z}^- (-1 \geq x)$$

we can proceed by universal generalization and take an arbitrary negative integer x , which by definition means $x < 0$, and since x is an integer, guarantees $x \leq -1 = g$, as required.

- (a) The quotient of any even number with any nonzero even number is even.

Solution:

The key word for the logical structure for this statement is “any”, which means that it is an **universal** statement, as we can see by translating it to symbols:

$$\forall e \in E \forall f \in E^{\neq 0} (e \text{ div } f \text{ is even})$$

where we define $E = \{x \in \mathbb{Z} \mid x \text{ is even}\}$ and $E^{\neq 0} = \{x \in \mathbb{Z}^{\neq 0} \mid x \text{ is even}\}$.

To prove an universal statement, the proof strategies we could use are universal generalization or structural induction (because the set of even numbers can be recursively defined by putting the element 0 in the set in the base case and then, in the recursive step, adding 2 or subtracting 2 to existing elements to get new elements) or proof by cases or proof by contradiction. Notice that proof by exhaustion is not possible because the domain is not finite.

To disprove an universal statement, main proof strategy we could use is to find a counterexample. We can also apply DeMorgan’s Laws to write the negation as an existential statement and use strategies (like finding witness or prove by cases) to prove it.

The statement is false, as we can see from the counterexample $e = 2$, since it is even and when we evaluate

$$\forall f \in E^{\neq 0} (2 \text{ div } f \text{ is even})$$

we can proceed again by counterexample $f = 2$, since it is even and nonzero and when we evaluate

$$e \text{ div } f = 1 \text{ which is not even}$$

Hence, we used $f = 2$ to disprove $\forall f \in E^{\neq 0} (2 \text{ div } f \text{ is even})$, and we used $e = 2$ to disprove $\forall e \in E \forall f \in E^{\neq 0} (e \text{ div } f \text{ is even})$

- (b) There are two odd numbers (not necessarily distinct) whose sum is even.

Solution: The main logical structure for this statement is that it is an **existential** statement, as we can see by translating it to symbols:

$$\exists x \in D \exists y \in D (x + y \text{ is even})$$

where we define $D = \{x \in \mathbb{Z} \mid x \text{ is odd}\}$.

To prove an existential statement, the main proof strategy we could use is to find a witness. Proof by cases and proof by contradiction could also be used, because both can be used to prove any statement (no matter its logical structure).

To disprove an existential statement, we would need to prove its negation, which (using DeMorgan’s Laws) can be written as a universal statement. Therefore, to disprove this statement the strategies we could use are universal generalization or structural induction (because the set of odd numbers can also be recursively defined) or proof by cases or proof by contradiction. Notice that proof by exhaustion is not possible because the domain is not finite.

The statement is **true**, as we can see from the witness $x = 1$, since it is an odd number and when we evaluate

$$\exists y \in D (1 + y \text{ is even})$$

we can proceed by finding a witness for y . Take $y = 1$, since it is an odd number and when we evaluate

$$1 + 1 = 2 \text{ is even}$$

Hence, we used $y = 1$ to prove $\exists y \in D (1 + y \text{ is even})$, and we used $x = 1$ to prove $\exists x \in D \exists y \in D (x + y \text{ is even})$.

- (c) The greatest common divisor of 5 and 23 is 1 and the greatest common divisor of 7 and 19 is 1.

Solution: The main logical structure for this statement is “and”, which means that it is a **conjunction**, as we can see by translating it to symbols:

$$(gcd((5, 23)) = 1) \wedge (gcd((7, 19)) = 1)$$

To prove a conjunction, the proof strategy we could use is to prove each of its components as a subgoal. To prove that $p \wedge q$ is true, have two subgoals: subgoal (1) prove p is true; and, subgoal (2) prove q is true. We could also use proof by cases or proof by contradiction. To disprove a conjunction, main proof strategy is to disprove one of its components. To prove that $p \wedge q$ is false, it's enough to prove that p is false. To prove that $p \wedge q$ is false, it's enough to prove that q is false.

The statement is **true**. We have two subgoals to prove:

(1) WTS: $(gcd((5, 23)) = 1)$. **Proof:** The positive factors of 5 are 1 and 5, and the positive factors of 23 are 1 and 23. Hence, the greatest common factor is 1.

(2) WTS: $(gcd((7, 19)) = 1)$. **Proof:** The positive factors of 7 are 1 and 7, and the positive factors of 19 are 1 and 19. Hence, the greatest common factor is 1.

Since we have proved both subgoals, the conjunction is true.

- (d) There are two positive integers greater than 20 that have the same greatest common division with 20.

Solution: The main logical structure for this statement is that it is an **existential** statement, as we can see by translating it to symbols:

$$\exists x \in \mathbb{Z}^+ \exists y \in \mathbb{Z}^+ (gcd((20, x)) = gcd((20, y)))$$

To prove an existential statement, the main proof strategy we could use is to find a witness. Proof by cases and proof by contradiction could also be used, because both can be used to prove any statement (no matter its logical structure).

To disprove an existential statement, we would need to prove its negation, which (using DeMorgan's Laws) can be written as a universal statement. Therefore, to disprove this statement the strategies we could use are universal generalization or structural induction

(because the set of odd numbers can also be recursively defined) or proof by cases or proof by contradiction. Notice that proof by exhaustion is not possible because the domain is not finite.

The statement is **true**, and the theorem we can use is “For any positive integers a, b , if a divides b then $\gcd((a, b)) = a$ ”. If we choose witness $x = 40$, since it is a positive integer and when we evaluate

$$\exists y \in \mathbb{Z}^+ (\gcd((20, 40)) = \gcd((20, y)))$$

we can first evaluate $\gcd((20, 40))$, and it equals 20 as 20 divides 40. We can then proceed by finding a witness for y . Take $y = 60$, since it is a positive integer and when we evaluate

$$\gcd((20, 60))$$

it equals 20 as 20 divides 60. So $\gcd((20, 40)) = \gcd((20, y)) = 20$.

Hence, we used $y = 60$ to prove $\exists y \in \mathbb{Z}^+ (\gcd((20, 40)) = \gcd((20, y)))$, and we used $x = 40$ to prove $\exists x \in \mathbb{Z}^+ \exists y \in \mathbb{Z}^+ (\gcd((20, x)) = \gcd((20, y)))$.